

Pratham Petwal

📍 Dehradun ✉ prathampetwal100@gmail.com ☎ 9634496058 📁 Portfolio in LinkedIn 📄 GitHub

Summary

- Software Engineer skilled in **Full-Stack Development**, backend engineering, and Applied AI. Built scalable applications using **React, Next.js, FastAPI, Node.js, MongoDB, PostgreSQL, and RabbitMQ**, while solving **2000+ DSA problems** and developing high-accuracy RAG systems.

Experience

Full Stack & LLM Applications Intern - AR Genie , Dehradun

June 2026 – July 2026

- Developed **LLM-powered RAG pipelines** using **FastAPI, Qdrant**, and multilingual embeddings for Japanese/English technical documents, processing over **10,000+ PDF pages**.
- Improved retrieval performance to approximately **80–90% accuracy** by optimizing chunking strategies, metadata schemas, reranking, and prompt engineering for complex technical queries.
- Developed a **Chatbot Evaluation Platform** using **FastAPI** and **React**, enabling automated multi-metric LLM evaluation and standardized validation of RAG chatbot responses.

Achievements

- **Competitive Programming Excellence:** Ranked in the **Top 10%** on LeetCode (Contest Rating: **1800**); **4th** on college leaderboard at GeeksforGeeks. [LeetCode](#) [GFG](#)

Education

Shivalik College Of Engineering – B.Tech in Computer Science & Engineering (AI & ML)

Aug 2023 – July 2027

- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Object-Oriented Programming

Skills

Programming Languages: C++, Python, JavaScript (ES6+), TypeScript, HTML5, CSS3

AI / RAG: RAG, Reranking, RAG & LLM-as-Judge Evaluation, Prompt Engineering, Vector DBs (Qdrant), PageIndex , ML , DL

Frontend: React.js, Next.js, Tailwind CSS, Responsive UI, State Management

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT & SESSION Authentication, API Integration

Databases: MongoDB, MongoDB Atlas, PostgreSQL, SQL, Qdrant

Developer Tools: Git, GitHub, Postman, VS Code, Chrome DevTools, Docker ,RabbitMQ ,AWS,CI & CD

Core Skills: Data Structures & Algorithms, Problem Solving, Performance Optimization, Secure Authentication, Scalable Web Applications, Clean Code

Projects

API Guard – API Gateway, Key Management & Real-Time Analytics Platform

[Live Demo](#) [Git Repo](#)

- Developed a full-stack **multi-tenant API gateway** using **React, Node.js/Express, MongoDB, PostgreSQL, and RabbitMQ** for scalable, reliable API management.
- Designed **RBAC**-based tenant isolation on a dual-database architecture, with **JWT** auth, bcrypt hashing, and a real-time analytics dashboard.

Campus Cart – College Marketplace Platform

[Live Demo](#) [Git Repo](#)

- Built a full-stack marketplace (**Next.js, Node.js, MongoDB, WebSockets**) for students to buy, sell, and bid on second-hand items, with secure auth via **Clerk** and real-time bid/listing updates.
- Integrated **Redux Toolkit** for state management and **Cloudinary** for optimized image delivery.

Knowledge Hub – Personal Knowledge Management Platform

[Live Demo](#) [Git Repo](#)

- Built a full-stack platform (**Next.js, FastAPI, MongoDB, Supabase**) to upload and manage files, images, videos, links, and articles.
- Developed secure REST APIs with **JWT**, debounced search, and pagination, reducing API calls by **30%**; built a responsive UI using **Tailwind CSS, React Query**, and **shadcn/ui**.